D 10681 **(Pages : 2)** **Name.....**

Reg. No.....

FIFTH SEMESTER U.G. DEGREE EXAMINATION, NOVEMBER 2021

(CBCSS—UG)

Physics/Applied Physics

PHY 5B 06/APH 5B 06—COMPUTATIONAL PHYSICS

(2019 Admissions)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A (Short Answer Type)

Answer at least **eight** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 24.

1. Identify the functional differences of the Compiler and the Interpreter in the context of high-level computer languages.
2. Write an algorithm to check whether a given number is odd or even.
3. List the different datatypes in Python.
4. Write a Python program to print the multiplication table of 9 having 20 rows using *while* loop.
5. What is a *tuple* ? How it is different from *list* ?
6. Discuss about the functional differences of *break* and *continue* statements in Python.
7. Write a Python program to plot the function $y = \sin x$ within the region 0 to 2π.
8. Write a Python program to create a 3 × 2 matrix having random numbers ranging between 0 and 1 as elements.
9. Explain the least squares curve fitting procedures.
10. Construct a forward difference table for the following data :

x	:	0	1	2	3	4
y	:	1	3	9	27	81

11. How the subintervals width is associated with the error in Trapezoidal rule and Simpson's 1/3 rule for numerical integration ?
12. Discuss the advantages of numerical methods over analytical methods.

(8 × 3 = 24 marks)

Turn over

Section B (Paragraph/Problem Type)

Answer at least **five** *questions.*
Each question carries 5 *marks.*
All questions can be attended.
Overall Ceiling 25.

13. Write a Python program to print the squares of the integer numbers within the range entered by the user.

14. Discuss about different built-in operations on *list* in Python with the help of examples.

15. Write a program to create two 3 × 3 matrices and add them.

16. The table below gives the temperature T (in °C) and length l (in mm) of a heated rod. If $l = a_0 + a_1 T$, find the best value for a_0 and a_1 :

T (in °C)	:	20	30	40	50	60	70
1 (in mm)	:	800.3	800.4	800.6	800.7	800.9	801.0

17. Using Newtons forward interpolation formula obtain y (2), given that, y (1) = 24, y (3) = 120, y (5) = 336, and y (7) = 720.

18. Explain the Bisection method for finding the solutions of algebraic equations.

19. Write a Python program to simulate the motion of a body dropped into a highly viscous medium.

(5 × 5 = 25 marks)

Section C (Essay Type)

Answer any **one** *question.*
The question carries 11 *marks.*

20. (a) Find y (0.2) for $y' = (x - y)/2$, y (0) = 1, with step length 0.1 using Runge-Kutta method.

    (b) Write a Python program to simulate a two-dimensional projectile motion using Euler's method in a table.

21. (a) Explain the Newton-Raphson method to find the roots of a function.

    (b) Write a Python program to simulate a freely falling body using Euler's method in a table.

(1 × 11 = 11 marks)